

Madhav Mishra

IITN, Waranga, Butibori, Nagpur – 441108
+91-9536068062
madhavamishra763@gmail.com

Portfolio: www.madhavamishra.me
LinkedIn: [linkedin.com/in/madhav-gfn/](https://www.linkedin.com/in/madhav-gfn/)
GitHub: github.com/madhav-gfn
LeetCode: leetcode.com/u/SmirkingCow/

EDUCATION

Indian Institute of Information Technology, Nagpur <i>B.Tech — Computer Science and Engineering (HCI & Gaming Technology); GPA: 7.93</i> Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, OOP, Algorithm Design, Human Computer Interaction, Software Engineering, Computer Vision Techniques, Computer Graphics	Nagpur, India July 2023 – June 2027
--	--

EXPERIENCE

Henwic Biomedics <i>Freelance Full Stack Development – Product Engineering</i> Developed an interactive biomedical product marketing website with React, TypeScript, and WebGL shaders; implemented scroll animations, particle effects, and lead capture integration with EmailJS and Google Analytics.	Remote / India May 2026 – June 2026
--	--

PROJECTS

TeacherJi: AI Learning Platform <i>FastAPI, React, LangGraph, Groq, FAISS, PostgreSQL, Redis</i> April 2026 - Orchestrated a 3-agent tutoring system with a FAISS-backed retrieval pipeline over NCERT textbooks, requiring precise state management among distributed agents to eliminate LLM hallucination. - Engineered the 3-step learning lifecycle backend; concept delivery, quiz generation, and adaptive feedback loops with data integrity spanning PostgreSQL profiles and Redis session state. - Deployed to Render and Vercel; validated sub-200ms end-to-end inference with Groq (OpenAI OSS 120B) under real network conditions.	
RAG-Powered Local Code Review Engine <i>ONNX, FAISS, llama.cpp, Tree-sitter, C++20, SQLite</i> July 2026 - Built a local RAG pipeline: Tree-sitter parses code into function/class-level units, a Qwen2.5-Coder ONNX model embeds them, and a FAISS HNSW index retrieves relevant context with content-hash incremental re-indexing. - Enforced grammar-constrained decoding (GBNF) at the LLM sampling layer to guarantee structurally valid JSON output, powering a deterministic severity-scoring engine. - Cut inference latency via per-file batched calls sharing a cached prompt KV state and a persistent-model daemon with idle-timeout auto-shutdown.	
AI-Native CRM with Campaign Automation <i>TypeScript, Node.js, PostgreSQL, Prisma, Next.js</i> June 2026 - Designed a 6-model PostgreSQL schema with Prisma ORM; constructed a chunked dispatch pipeline(50 msg/batch, <code>Promise.allSettled</code>) and serializable webhook processor with status-rank guards against out-of-order events. - Integrated Google Gemini 2.0 Flash at three AI features: NL-to-filter segment creation with JSON enforcement, a RAG-based Campaign Copilot generating channel and message copy, and a post-campaign insight engine. - Programmed 15+ REST endpoints covering customer, segment, campaign, and webhook modules; wired a cron-driven scheduler, a decoupled channel-simulator microservice, and a Next.js 16 / Tailwind CSS 4 dashboard tracking SENT → DELIVERED → OPENED → CLICKED funnels.	
JSON-Driven Educational Game Engine Platform <i>TypeScript, React, Express, SQLite, Zod</i> Jan 2025 – Present - Architected a zero-code game engine for students with Zod-validated JSON configs, enabling 12+ game types through pluggable renderers without modifying engine core. - Incorporated an AI generation pipeline transforming prompts into valid JSON configs, auto-validating against 8+ schema constraints and generating level content for instant deployment. - Established a leaderboard API with 3-tier tie-breaking over SQLite WAL, handling 1000+ submissions/second with admin dashboard for real-time analytics and CRUD operations.	

SKILLS

Languages	C++, Python, TypeScript, JavaScript, C, C#
Systems, Web & Backend	RESTful APIs, Node.js, Express.js, React, Tailwind CSS, Zustand, FastAPI, PostgreSQL, MySQL, MongoDB, Redis, SQL, Next.js, Prisma
AI & LLM	LangGraph, AI Agents, RAG (FAISS), Groq API, LangChain, Hugging Face, Agentic Workflows
Cloud & Infra	Google Cloud, Oracle Cloud, Prometheus, Grafana, VirtualBox
Dev Tools	Git, GitHub, Postman, Jira, Agile/Scrum

ACHIEVEMENTS & INTERESTS

- Secured **1st Place** /20+ teams: TECHNEX’25 Game Jam, IIT BHU (2025), deception-driven tower defense
- Achieved **1st Prize** /30+ teams, Tantarfiesta’25 Designathon, non linear emotion based journaling app design
- Placed 3rd Runner-Up — Exergy Case Study Challenge, IIT Kanpur (2025)
- Interests:** Software Development, Scalable Systems, HCI, AI/LLM Engineering

CERTIFICATIONS

Fundamentals of Accelerated Computing with CUDA Python View Credential	April 2026
Complete Generative AI Course with LangChain and HuggingFace View Credential	Jan–April 2026
Oracle Cloud Infrastructure 2025 Certified Foundations Associate View Credential	Nov 2025